Chapter 5: Errors, Power, and Effect Size

Statistics for Environmental Science

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> *It is better to be roughly right than precisely wrong.*
>
> Attributed to John Maynard Keynes

Chapter 4 ended with a hypothesis test that either rejects H_0 or fails to reject it.

This chapter first asks how often that decision is wrong, and in which direction. It then asks how large the effect is and whether the study had enough data to detect it. It closes with how to report a result.

Errors in hypothesis testing

Sometimes, even when we apply the test correctly, sampling variability can lead us to the wrong conclusion.

- **Type I error (α):** Rejecting H_0 when it is actually true. This is like a smoke alarm going off when there's no fire: a false alarm. By convention, α is often set at 0.05 (5%).
- **Type II error (β):** Failing to reject H_0 when it is actually false. This is like a smoke alarm staying silent when there *is* a fire: a missed signal.

Decision	If the null hypothesis is True	If the null hypothesis is False
Reject H_0	Type I error (prob = α)	Correct (prob = 1 - β)
Fail to reject H_0	Correct (prob = 1 - α)	Type II error (prob = β)

For a fixed sample size and effect size, lowering α raises β, because a stricter standard for rejecting H_0 makes it easier to miss a real difference.

Figure 1 recaps the test from Chapter 4. The distance between the two sampling distributions, in standard errors, decides whether we reject H_0.

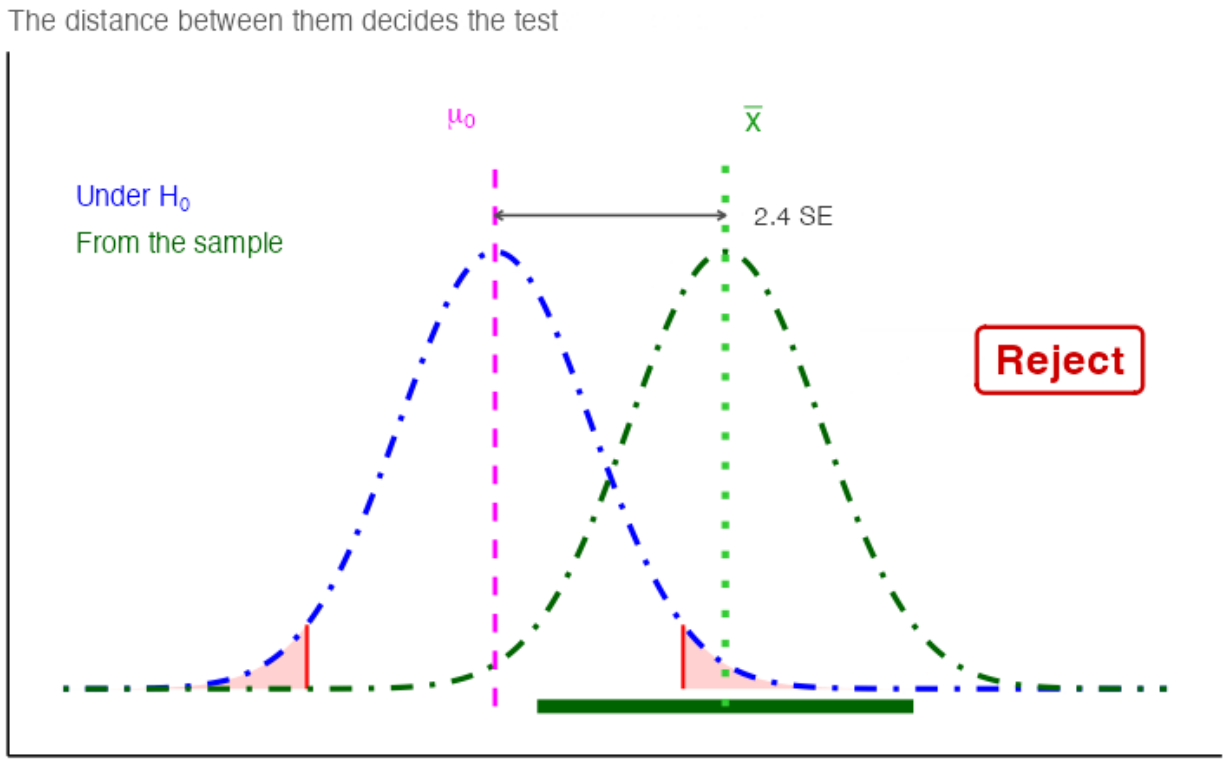


Figure 1: The two sampling distributions of the sample mean from Chapter 4. The distance between them, in standard errors, decides the test: past 1.96, reject H_0. (Final frame. The animated version is in the web edition of this book.)

- In Figure 2, the true population mean (orange) is the same as μ_0. By chance, however, the sample mean lands far away, and the CI excludes μ_0. The test rejects H_0 even though it is true: a Type I error.

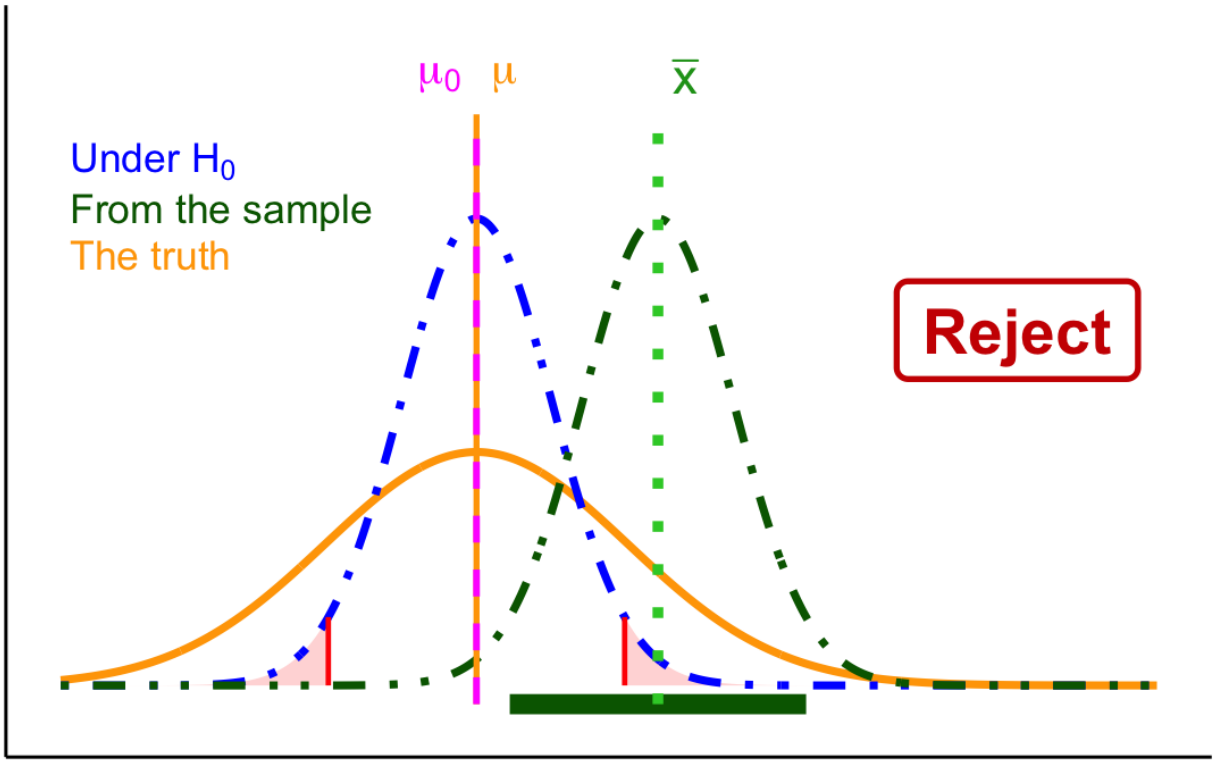


Figure 2: Type I error: H_0 is true ($\mu = \mu_0$), but the sample leads us to reject H_0.

- In Figure 3, the true population mean (orange) is far from μ_0, but the particular sample looks close to μ_0. The CI includes μ_0, so the test fails to reject when it should reject the null: a Type II error.

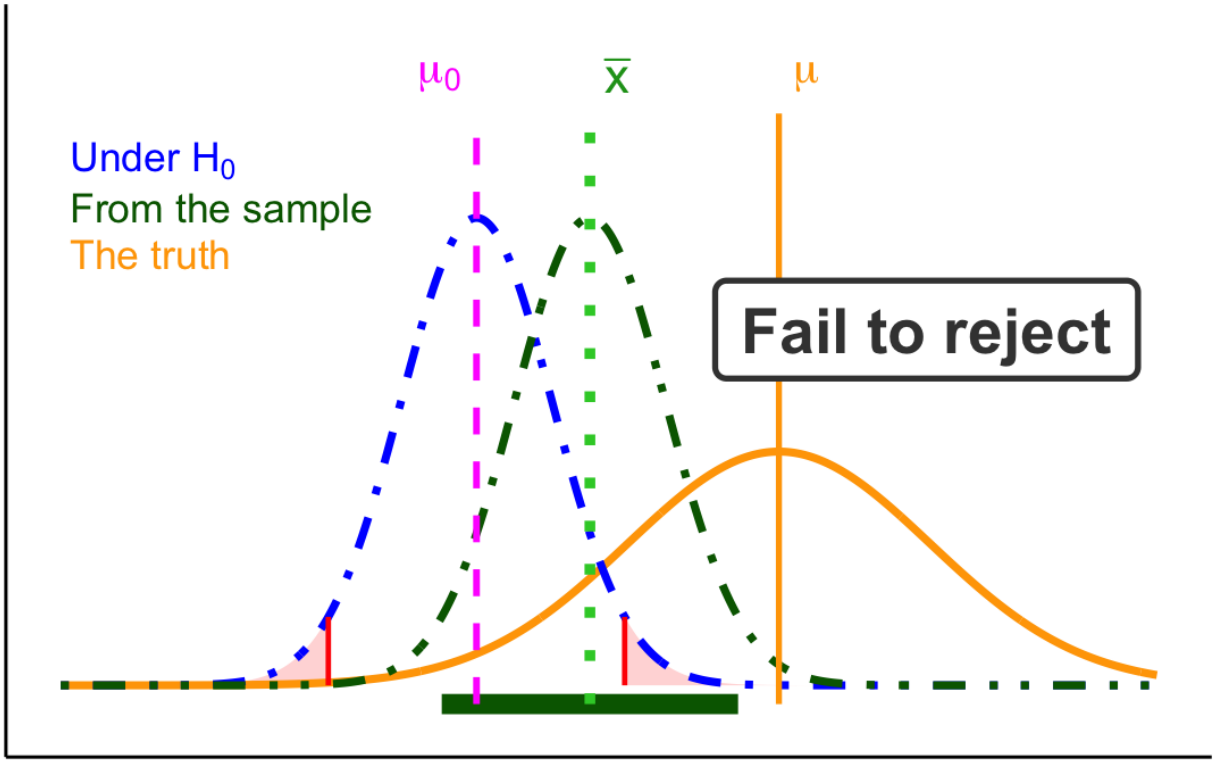


Figure 3: Type II error: H_0 is false (true population mean differs from μ_0), but the sample leads us to fail to reject H_0.

Those two figures show what a wrong conclusion looks like for one unlucky sample. They do not show how *often* it happens. The green curve in them is built from that one sample, centered on its $\bar{x}$. To see how often an error happens, center the sampling distribution on a possible truth instead. That shows where the sample mean lands across all the samples we could have drawn.

Figure 4 does that. The blue curve is the sampling distribution of the sample mean under H_0, the one that sets the critical values. The orange curve is the sampling distribution of the sample mean if $\mu = \mu_a$. Chapter 4 drew the possible true populations under H_a as a family of curves on either side of μ_0, and μ_a is the center of one of them. Because we never know the true μ, we choose a value for μ_a and ask how the test would perform if that value were true. The farther μ_a is from μ_0, the less likely the test is to miss the difference.

- α is the red area in the tails of the blue curve, past the critical values. If H_0 is true, that is how often the sample mean lands in the rejection region anyway. In that case the blue curve is the sampling distribution of the sample mean, so α needs only the blue curve. That is why the top panel has no orange curve. The orange curve enters in the bottom panel, for β.
- β is the gray area under the orange curve *between* the critical values. If $\mu = \mu_a$, that is how often the sample mean fails to reach the rejection region and a real difference goes undetected.
- Power, $1 - \beta$, is the rest of the orange curve, the part past the critical values.

In Figure 4, μ_a is 2.5 standard errors above μ_0, which leaves $\beta = 0.29$ and power $= 0.71$. This design misses a real difference of that size in almost three tries out of ten. Slide the orange curve further right, meaning a larger real difference, and the gray area shrinks. Move the red lines outward, meaning a smaller α, and the gray area grows.

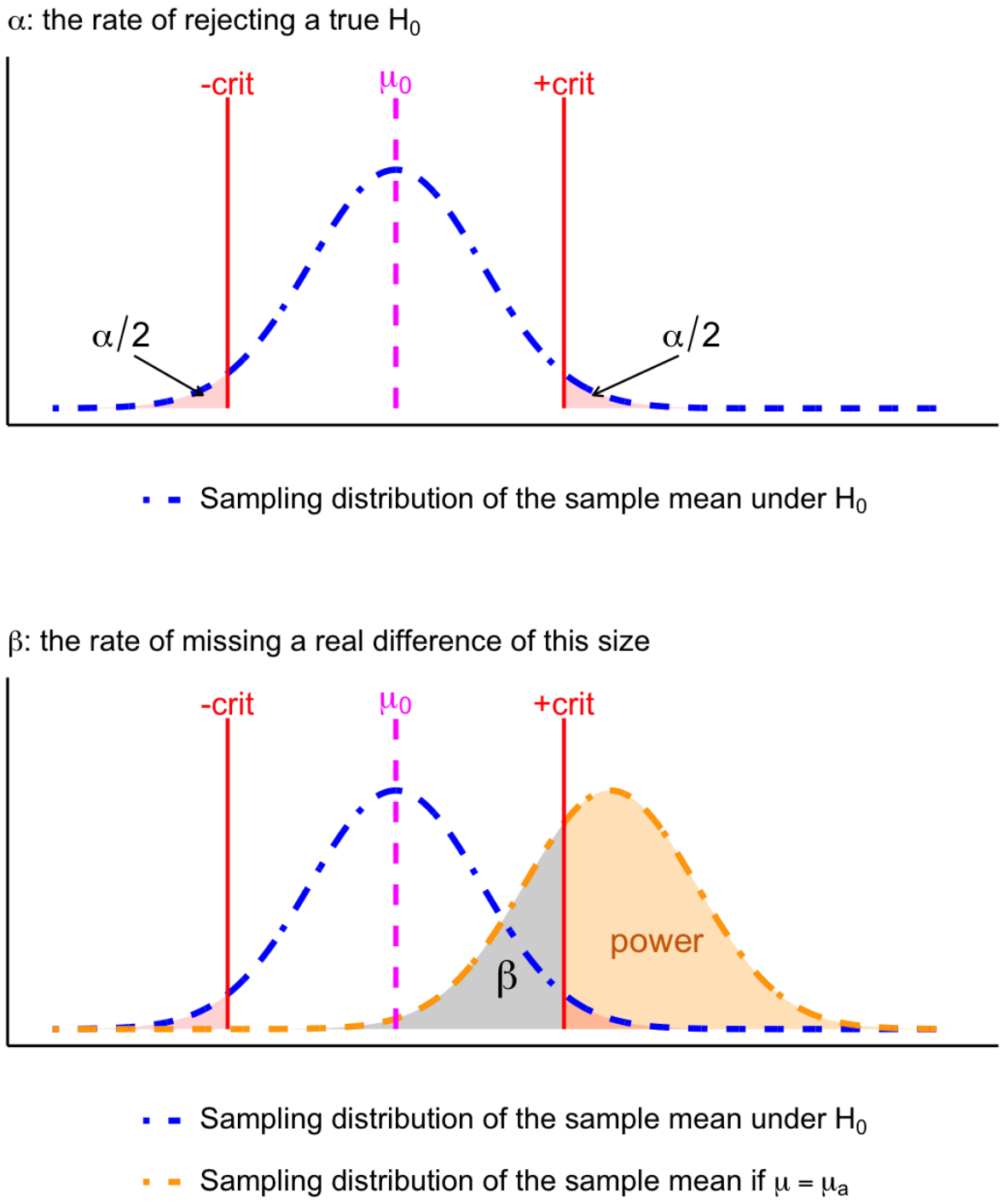


Figure 4: The two error rates, drawn on sampling distributions. Top: α is the red area in the tails of the curve under H_0. Bottom: adding the curve for one specific alternative value μ_a, β is the gray area under it between the critical values, and power is the orange area past the right critical value.

Effect of sample size on errors

Sample size mainly affects Type II error. With a fixed α and correct assumptions, the long-run Type I error rate is α for any n. With small n, the sampling distribution is wide, true effects are harder to detect, and β increases (power decreases). As n grows, the sampling distribution tightens, β falls, and power rises. Apparent changes in the Type I rate at small n usually reflect assumption violations (e.g., non-normality, poor variance estimates, multiple looks), not n itself.

Figure 5 shows this with the same μ_0, the same true μ, and the same unlucky sample, in which $\bar{x}$ lands 2.5 standard errors below μ. With the smaller sample, the confidence interval still catches μ_0 and we fail to reject. With four times the sample, the standard error halves, the interval clears μ_0, and we reject H_0.

Choosing the tail after seeing the data. Chapter 4 said to choose the tail before you see the data, because picking it afterward makes you reject more often than your stated α implies. A one-tailed test at $\alpha = .05$ puts the whole 5% in one tail, so if you pick the tail *after* seeing which way the data went, you have effectively run a two-tailed test at $\alpha = .10$ and doubled your Type I error rate. A one-tailed test with its direction chosen in advance keeps the Type I error rate at α.

How to interpret *p*-values

A p-value is the probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the one we observed. We reject at level α when $p < \alpha$.

Tails. Count only the tail or tails that H_a points to: both tails for a two-tailed test, one tail for a one-tailed test.

Warning: Common mistakes to avoid

1. p is *not* the probability that H_0 is true.
2. A tiny p is not "super significant." It means only that the data would be very unlikely if H_0 were true. The effect itself can still be tiny and not worth acting on, especially with a large n. Report an effect size and a CI so a reader can judge whether it matters.
3. With small n, true effects can yield large p (low power).

Beyond the 0.05 cutoff

People who act on a result, such as scientists, managers, and policymakers, need to know how large an effect is and how likely the study was to detect it. A decision at $\alpha = 0.05$ answers neither question. Effect size and power answer them.

Working definitions

- **Statistical significance:** the result is unlikely under H_0, at your chosen α.
- **Practical significance:** the effect is large enough to change what someone does about it. This is a judgment about the world, not a calculation. With a large enough n, a result can be statistically significant and practically trivial. With too small an n, a result can be practically important and still not statistically significant.
- **Effect size:** how large the effect is, in the study's units or in standard deviations.
- **Power ($1 - \beta$):** the probability your design will detect a real effect of a specified size at your chosen α.

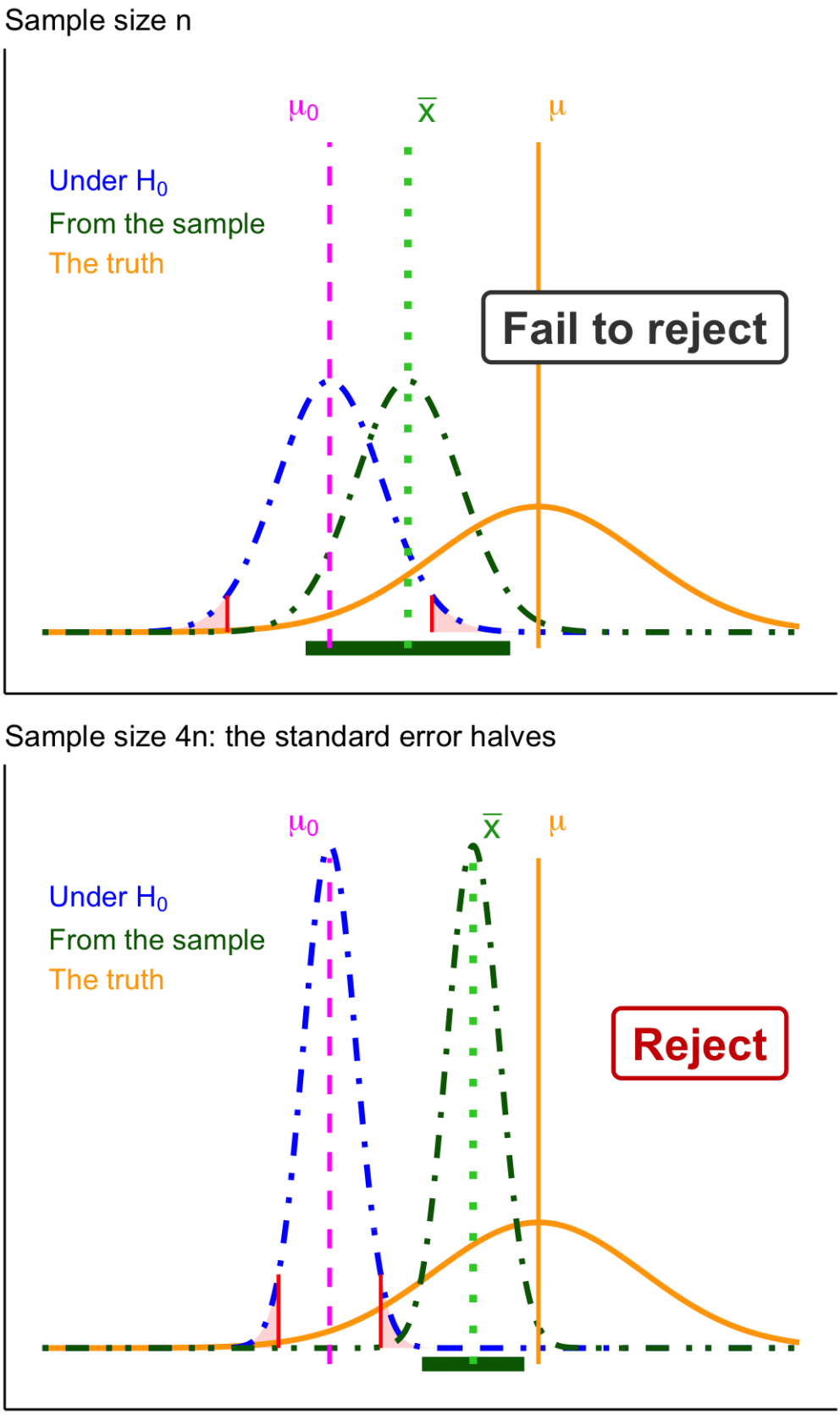


Figure 5: The Type II situation at two sample sizes, with the same μ_0, true μ and population. In both, $\bar{x}$ lands 2.5 standard errors below μ. Top: the confidence interval catches μ_0, so we fail to reject. Bottom: with four times the sample, the standard error halves; the same bad luck leaves the interval clear of μ_0, and we reject.

Plan effect size and power before you sample. Decide what effect is *meaningful* for your context (e.g., a 0.3 mg/L nitrate reduction) and size your study to have adequate power to detect that effect.

Effect size

Effect size comes in two kinds of units. You can express it in original units, which carry practical meaning, or in standardized units, which let you compare across scales.

Practical (original units). Use the scale stakeholders recognize.

- Exams: "Treatment improved scores by 5 percentage points" (half a letter grade).
- Time-on-task: "New training reduces practice time by 1,000 hours out of 10,000" (a 10% reduction).

These statements are easy to interpret, but they are hard to compare across contexts or measures.

Standardized (unitless). When units are unfamiliar or scales differ, standardize by typical variability. The most common standardized effect size metric is **Cohen's *d*** (Cohen 1988). It divides the raw difference, Δ, by the population standard deviation:

$$\text{Cohen's } d = \frac{\Delta}{\sigma}$$

Δ is $\mu_1 - \mu_2$ when comparing two groups, or $\mu - \mu_0$ when testing one mean against a reference value. Now "how big?" is expressed in SD units.

Cohen's d has the same numerator as z but a different denominator. Chapter 4 gave the general shape, effect over error, and filled it in with the z:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \text{versus} \qquad \text{Cohen's } d = \frac{\mu_1 - \mu_2}{\sigma}$$

Both numerators are the effect. The test statistic divides by the standard error, which shrinks as n grows, so z gets larger with more data even when the effect stays exactly the same. Cohen's d divides by the standard deviation, which does not shrink with n, so Cohen's d reports the size of the effect itself.

A large test statistic therefore does not mean a large effect, and the two numbers answer different questions. With enough data, a trivial difference can produce a z as large as you like, while Cohen's d stays small. The t in Chapter 6 is the same ratio with an estimated s in place of a known σ, so the same holds for t.

Environmental example (biodiversity index). Suppose an index increases from 3 to 4 after reforestation.

- In raw units: +1 index point.
- In standardized terms:
  - If the SD across sites is 1.0, then Cohen's $d = 1.0$: a large shift.
  - If the SD is 10, then Cohen's $d = 0.10$: a small shift.

Planning uses Δ. To plan a study, Δ is enough: it is the size of the effect you expect to see, in the units of the study, and there is no need to standardize it. Cohen's d standardizes it so that effect sizes can be compared across studies.

Notes for practice

- In real data, σ is unknown; estimate it (pooled SD for two groups, with small-sample corrections as needed).
- Heuristics (context matters): Cohen's $d \approx 0.2$ (small), 0.5 (medium), 0.8 (large).

- A "small" Cohen's d can still be important (e.g., small reductions in lead or $PM_{2.5}$ can matter for health).
- Report a 95% CI with each effect size to show how precisely it was estimated.

Suppose we're measuring final exam performance (percent correct). The class mean is 65% with a standard deviation of 5%. Group A is a control, while Group B receives some instructional treatment.

Figure 6 shows four possible scenarios:

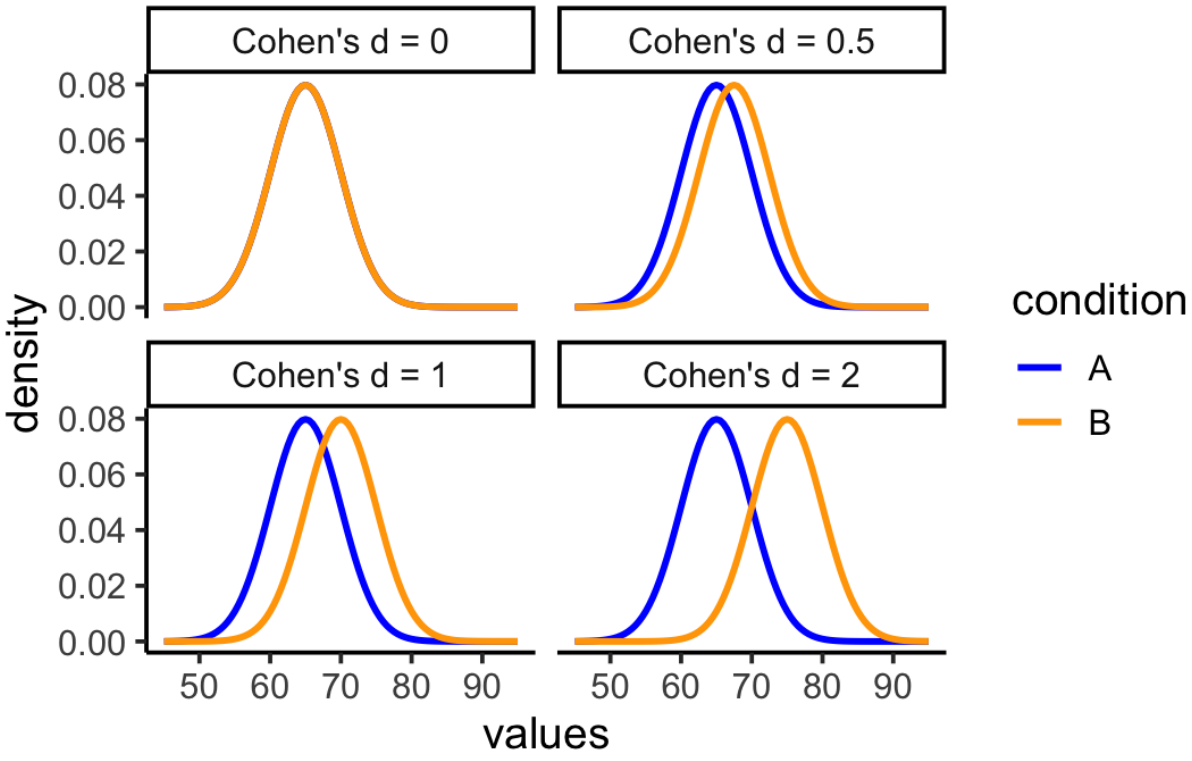


Figure 6: Same idea with concrete scores (mean 65, $sd = 5$): treatment shifts correspond to Cohen's $d = 0$, 0.5, 1, 2.

- **Panel 1 (Cohen's $d = 0$).** No effect. Both groups sample from the same distribution (mean = 65).
- **Panel 2 (Cohen's $d = 0.5$).** Group B's mean shifts upward by 2.5 points (67.5).
- **Panel 3 (Cohen's $d = 1$).** A 5-point shift (70).
- **Panel 4 (Cohen's $d = 2$).** A large 10-point shift (75).

In Figure 6, a half-SD shift (Cohen's $d = 0.5$, 2.5 points) leaves the two curves mostly overlapping, and a two-SD shift (Cohen's $d = 2$, 10 points) separates them. In practice, we rarely know beforehand how big a shift to expect.

Power

Effects vary in size, and study designs vary in how reliably they detect an effect of a given size. That reliability is the design's statistical power. A design with low power will usually fail to reject H_0 even when a real effect exists.

Power rises when:

1. Effect size is larger.
2. Sample size (n) is larger (smaller SE).
3. The test is less conservative (larger α).
4. Measurement noise is lower (smaller SD).

For planning, the practical question is how large n must be to reach 80% power, with α fixed at 0.05, for the smallest effect that matters. Figure 8 answers that question for one effect size.

Simulation walkthrough

Let's see how design choices affect power with a simple simulation. We simulate two groups of $n = 10$ each. We draw Group A (control) from a normal distribution with mean 10 and $SD = 5$, and Group B (treatment) from one with mean 12.5, a shift of 2.5 points, or Cohen's $d = 0.5$, a medium effect.

If we simulate this experiment 1,000 times and run an independent-samples t-test each time, how often do we reject H_0 at $\alpha = 0.05$?

```
set.seed(2025)
p <- numeric(1000)
for(i in 1:1000){
  A <- rnorm(10, 10, 5)
  B <- rnorm(10, 12.5, 5)
  p[i] <- t.test(A,B,var.equal=TRUE)$p.value
}
power_n10 <- mean(p < 0.05)
```

The proportion of simulated experiments that reject H_0 estimates the design's power, the probability that the test detects a real difference of this size. Here it is about 21%. With $n = 10$ per group, the test misses this medium-sized effect in most experiments.

How power changes

1. **Larger n.** Doubling to $n = 20$ per group reduces sampling error:

```
p <- numeric(1000)
for(i in 1:1000){
  A <- rnorm(20, 10, 5)
  B <- rnorm(20, 12.5, 5)
  p[i] <- t.test(A,B,var.equal=TRUE)$p.value
}
power_n20 <- mean(p < 0.05)
```

That increases power to about 33%, up from about 21% at $n = 10$.

2. **Smaller α.** Making the test more conservative (e.g., $\alpha = 0.01$) lowers power, because fewer outcomes count as "significant". Using the same $n = 20$ simulations, we just change the cutoff:

```
power_n20_strict <- mean(p < 0.01)
```

Power falls to about 12%.

3. **Larger effect.** If the treatment produces a big shift (say 2 standard deviations), the same design at the same $\alpha = 0.01$ has near-perfect power:

```r
p <- numeric(1000)
for(i in 1:1000){
  A <- rnorm(20, 10, 5)
  B <- rnorm(20, 20, 5)   # 2SD effect
  p[i] <- t.test(A,B,var.equal=TRUE)$p.value
}
power_big <- mean(p < 0.01)
```

Power is now about 100%. A 2-SD difference is detected in almost every experiment, even with 20 samples per group and $\alpha = 0.01$.

Power curves

A design's power depends on how large the true effect is. A power curve plots the power of one fixed design against a range of effect sizes. Figure 7 fixes the design at a two-sample t-test with $n = 10$ per group and $\alpha = 0.05$, and plots power against the true effect size, Cohen's d.

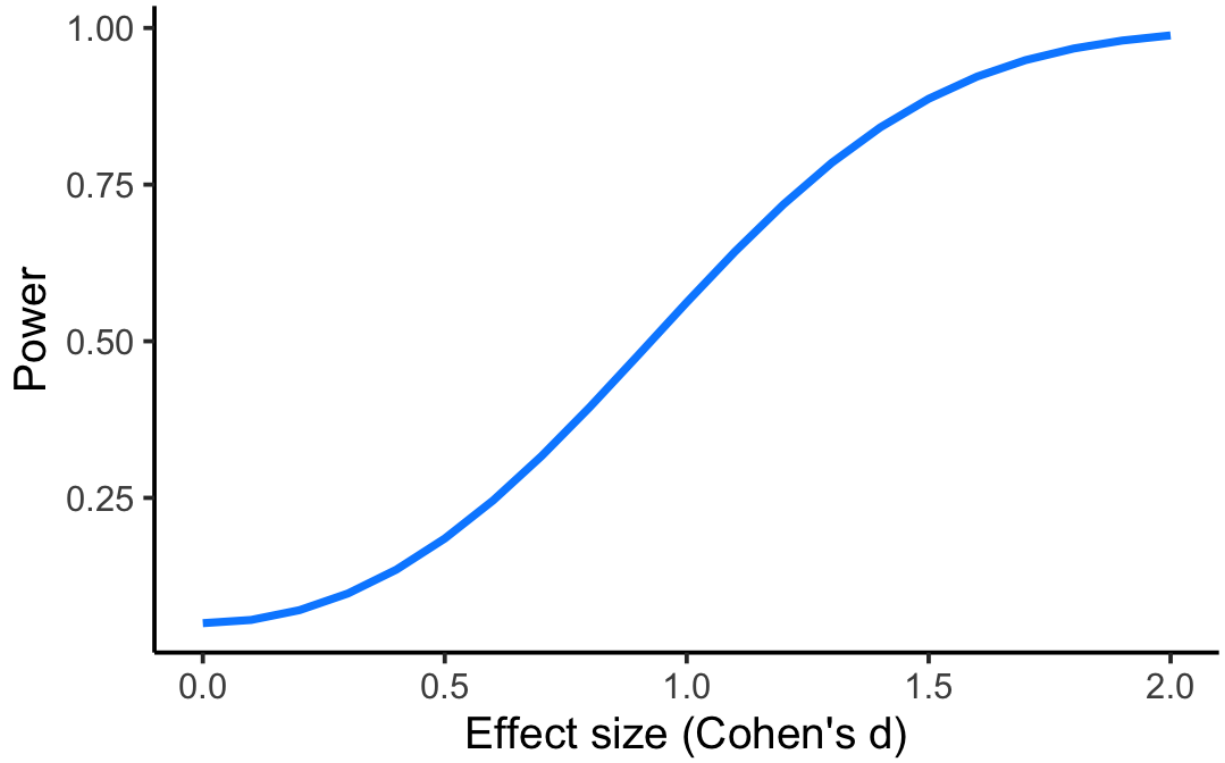


Figure 7: Power vs. effect size (Cohen's d) for a two-sample t-test with $n = 10$ per group and $\alpha = 0.05$.

With 10 per group, this design detects a medium effect (Cohen's $d = 0.5$) only about 18% of the time, a larger effect (Cohen's $d = 0.8$) about 40% of the time, and a very large effect (Cohen's $d = 2$) almost every time (about 99%). With $n = 10$ per group, only large effects are detected reliably.

The simulation above estimated power at Cohen's $d = 0.5$ as about 21%. The two numbers differ because 1,000 simulated experiments give an estimate, while the curve is computed exactly.

The second curve holds the effect size fixed and varies the sample size. Suppose prior research suggests the effect is small (Cohen's $d = 0.2$). Figure 8 plots power against sample size for Cohen's $d = 0.2$, with a dashed line at 80% power, a common planning target.

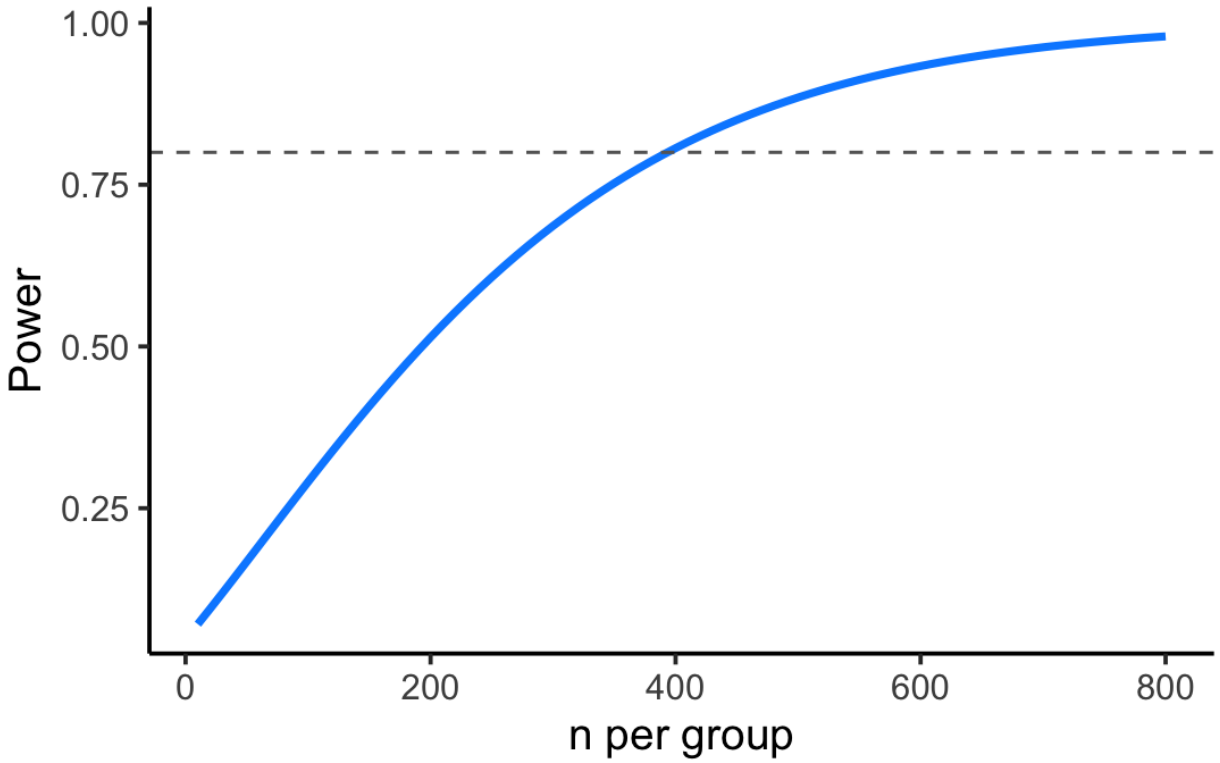


Figure 8: Power vs. sample size for detecting Cohen's $d = 0.2$ with a two-sample t-test at $\alpha = 0.05$; dashed line marks 0.80 power.

At 10 per group, power to detect Cohen's $d = 0.2$ is about 7%. Reaching 80% power takes about 394 per group. At that sample size, 20% of studies would still fail to reject H_0 for a real effect of Cohen's $d = 0.2$.

Power analysis is a planning tool. A power target such as 0.8 is calculated for a specific effect size and standard deviation, so report both with it.

Planning your design

Planning a study means working through the following, in order:

- Define a meaningful effect, Δ: the smallest difference from μ_0 that would change what you do. State it in the units of the measurement.
- Choose α (often 0.05; justify if different).
- Estimate variability, σ (pilot data, literature, or historical monitoring). With Δ and σ you have Cohen's d, which is what a power calculation needs.
- Compute n for $\geq 80\%$ power (or explain tradeoffs if you can't reach it).
- Pre-specify one- vs two-tailed before data collection.

This sequence is a **formal power analysis**. It applies when you are about to collect data and can still choose n, for example when deciding how many plots to establish or how many samples a permit application

commits you to. When the data already exist, as in your course project, n is fixed. Report the effect and its confidence interval instead.

> Warning: Post-hoc power
>
> Do not compute power after the fact for the effect you observed. This **post-hoc power** is a restatement of the p-value, so a non-significant result always has low post-hoc power. Power for a difference chosen in advance, such as the smallest effect that matters, is fine to report.

With 10 per group, only effects of about Cohen's $d = 1.3$ or larger reach 80% power, and detecting Cohen's $d = 0.2$ with 80% power takes about 394 per group. Plan the design so it can reliably detect the effects it is meant to measure.

Two cautions

- **A significant result from a low-power study is less trustworthy.** Underpowered studies rarely replicate. A design with only 30% power to detect an expected effect might produce a significant result by luck, but most replications will not. When a small study does reach significance, its estimated effect size tends to be too large. When the null is true, that large estimate makes a false positive look convincing.
- **Huge n finds tiny effects.** With very large samples, trivial differences can register as statistically significant. For example, a satellite image can contain millions of pixels, so a difference of a fraction of a percent in greenness between two land-cover types can come out statistically significant, even though a difference that small would not change how either land cover is managed. Report an effect size and a confidence interval alongside each p-value so a reader can judge whether a difference matters.

Figure 9 shows the first caution. When the null hypothesis is true, 5% of tests still give $p < 0.05$. In small samples, those false positives come with large estimated effect sizes, even though the true effect is zero.

Reporting results

A results statement is one sentence that states the finding in plain language and ends with the key statistics, in a fixed order, in parentheses.

Template (two-sided test): *Plain-language finding. (test, statistic, df or n, value of statistic, p-value, effect size/CI if relevant).*

What to include (in order):

1. Test name (e.g., one-sample z, two-sample t, χ^2, regression).
2. Test statistic and its degrees of freedom (or sample size for z).
3. Exact p (report $p < 0.001$ when very small).
4. An effect size and/or a 95% CI when available.

Examples

- **One-sample z.** The average mercury level within 10 km of the smelter was higher than the legal limit ($z = 2.85$, $n = 32$, $p = 0.004$).
- **One-sample t with CI.** Mean nitrate exceeded the target by 0.41 mg/L ($t(31) = 2.6$, $p = 0.014$, 95% CI [0.09, 0.73] mg/L).
- **Two-sample t with effect size.** Downstream sites had higher turbidity than upstream sites ($t(58) = 3.1$, $p = 0.003$, Cohen's $d = 0.80$).

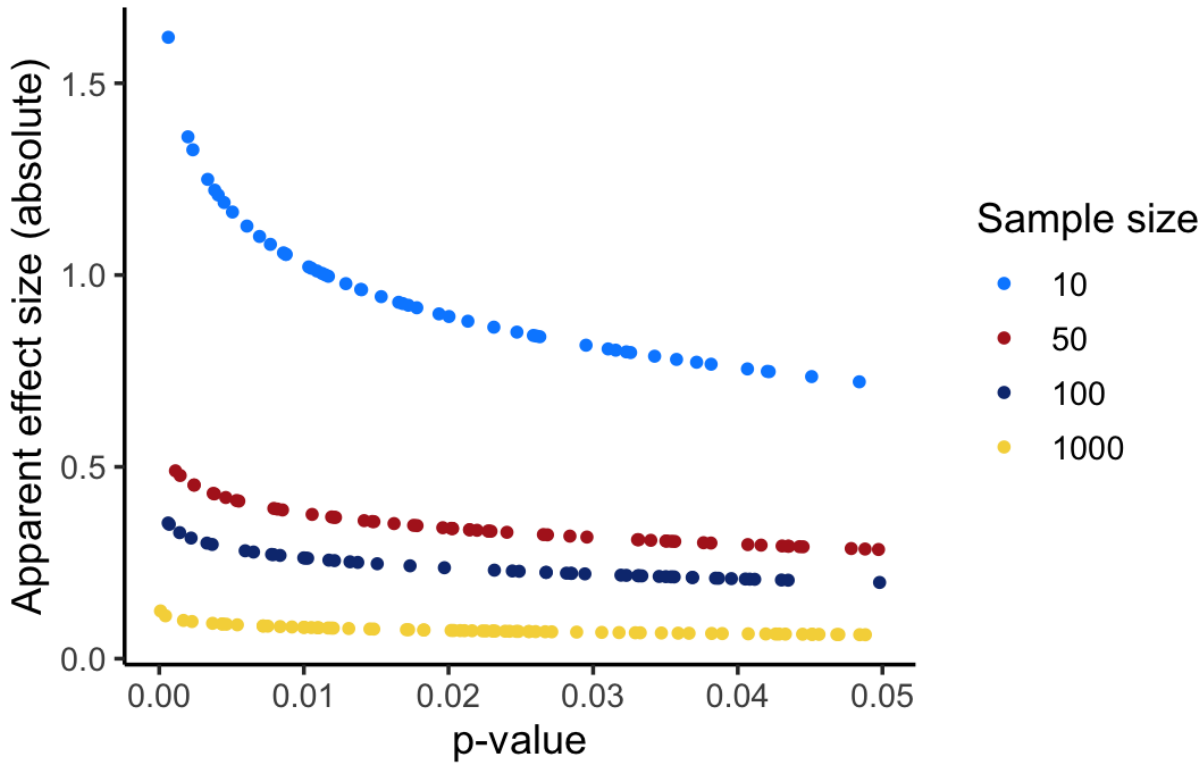


Figure 9: Apparent effect size among Type I errors ($p < 0.05$) under the null, by sample size, for a one-sample t-test. Effect sizes are shown as absolute values, since a false positive is equally likely in either direction and the sign carries no information here. Small n inflates apparent effects.

- **Not significant (be explicit).** Mean canopy temperature did not differ from the baseline ($t(47) = 1.2$, $p = 0.24$); we fail to reject H_0.

Rounding & style

- Round statistics to 2 decimals; p to 3 decimals (use $p < 0.001$ when needed).
- Use "fail to reject H_0" (not "accept").
- Match tail to hypothesis in prose (e.g., "higher than," "lower than," or "different from").

Do / Don't

- Do pair the sentence with a short, substantive takeaway (what it means for the question).
- Do include a CI or effect size when it aids interpretation.
- Don't restate methods; don't claim proof; don't omit units.

Chapter Summary

Why it matters. A hypothesis test ends in reject or fail to reject, and that decision can be wrong in two ways. Effect size tells you how large the difference is, and power tells you whether the study could have detected it.

Core ideas

- **Two ways to be wrong.** A Type I error rejects a true H_0, at a rate you set with α. A Type II error fails to reject a false one. For a fixed design, lowering α trades fewer Type I errors for more Type II errors. $\alpha = .05$ is a widely used convention for where to make that trade.
- **What a p-value measures.** It is the probability of data at least as extreme as ours if H_0 were true. It is not the probability that H_0 is true, and it says nothing about how large an effect is.
- **Significance and effect size answer different questions.** A significant result says the data would be unlikely if H_0 were true. With a large enough n, a trivially small difference is significant. Report an effect size and a confidence interval next to every p-value so a reader can judge whether the result matters.
- **Effect size in two kinds of units.** The same difference can be written in original units, Δ, which carry practical meaning for the people who act on your result, or in standard deviations, Cohen's $d = \Delta/\sigma$, which makes results comparable across studies and scales. Report both when you can, and plan in Δ, because the smallest difference worth detecting is a judgment made in real units.
- **Power is a design decision.** Power depends on n, effect size, α, and variability together. An underpowered study usually misses real effects, and when it does reach significance, the effect it reports tends to be too large. Run a power analysis before collecting data, while n can still be chosen.
- **Report a result as a sentence.** State the direction in words, then give the test, the statistic with its df or n, the p-value, and an effect size or confidence interval. "Significant" on its own tells a reader nothing about direction, magnitude, or precision.

Cohen, J. 1988. *Statistical Power Analysis for the Behavioral Sciences*. Second. Lawrence Erlbaum.